PASSWORD REUSABILITY : USER BEHAVIOR, SECURITY RISKS AND MITIGATION STRATEGIES

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Abstract

One of the main problems in modern cybersecurity is still the reuse of passwords. Despite being fully aware of the dangers, users frequently reuse passwords across many platforms in an attempt to prevent mental fatigue. This paper looks at the behavioral aspects that contribute to pass word reusability, the security risks that arise, especially from password stuffing attacks, and the performance of current technical countermeasures. Although user education is still crucial, we think that systemic reforms toward passwordless authentication and automated credential management are required to reduce this danger.

Keywords: Password; Reusability; Authentication, Cybersecurity; Behavior

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1 Introduction

The key aspect of the problem is the relationship between security and user experience in password systems. It discusses how password standards impact both the degree of protection against cyberattacks and user convenience. The main issue is the trade-off between security and usability. Stronger password requirements, including lengthy, complex passwords that are frequently changed, make users more vulnerable to hacking and phishing attacks. Strict rules, however, often lead to unintended user behavior, such writing down passwords, using them for several accounts, or choosing patterns that are predictable. These weaknesses threaten the exact security that those regulations were meant to protect. This happened because one of the most frequent attack methods is password leaks. Even the most advanced protections are useless if users cannot handle their credentials. In addition, its impact on the economy may lead to data breaches that cost companies millions of ringgit in reimbursement, legal liability, and reputational damage. Studying this topic is crucial to responding to evolving cyberthreats and developing policies that improve protection and usability while ensuring user compliance while maintaining security.

2 Problem Statement

The Root Problem

Passwords continue to be the most popular authentication technique, according to studies, but users find it difficult to comply with complicated criteria, which encourages unsafe behaviors like reuse or weak patterns. Strong security regulations and human usability clash, which frequently compromises security. Current strategies, including regular password changes or strict complexity guidelines, have downsides because they irritate users and fail to prevent current threats like phishing or credential stuffing. To ensure that security and usability are balanced, a review of the literature is required to find gaps, assess effectiveness, and investigate new approaches like multi-factor authentication or passwordless systems.

Research Gaps

Through analyzing multiple research works, several research gaps can be identified:

1. Small sample numbers and insufficient study verification methods: For instance, Kablo et al. (2024) only spoke with 11 participants when looking at security audit interfaces in password managers. Although their findings provide useful insights into usability and adoption, the small sample limits generalizability and highlights the necessity for larger-scale experimental validation to verify whether these outcomes hold true across broader groups.

2. Restricted geographic scope: Fernando et al. (2023) investigated password usage and security breaches, however they primarily used studies from specific regions, such as Sri Lanka and Europe. Due to the lack of global perspectives in this narrow emphasis, user behaviors and usability difficulties in various cultural and geographic contexts continue to be understudied. Determining the generalizability of findings might benefit from extensive worldwide research.

3. Absence of technique comparisons: Wasfi & Stone (2023) noted that there isn't a comprehensive side-by-side examination to assess the usability, memorability, and attack resistance of textual, passphrase, mnemonic, and recognition-based password schemes. Without such comparison studies, it is difficult to identify which approaches provide the best security and usability balance for different user groups.

3 Research Questions, Hypotheses and Objectives

Research Questions

1. What approaches have been used by researchers to improve password usability while maintaining strong security?
2. What are the major findings reported in existing studies regarding user behavior and password management?
3. What limitations exist in current password usability approaches, and how do they affect overall cybersecurity?
4. What research gaps and future opportunities can be identified to design more effective and user-friendly authentication systems?

Hypotheses

1. H1: Approaches that integrate password managers and multi-factor authentication significantly improve usability while maintaining strong security compared to traditional password-only systems.
2. H2: Users tend to prioritize convenience over security, leading to widespread password reuse and weak password choices, despite awareness of cybersecurity risks.
3. H3: Current usability-focused password policies, such as frequent mandatory changes, complex composition rules, inadvertently increase password reuse and reduce memorability, thereby weakening overall cybersecurity.
4. H4: Incorporating explainable password audit tools and adaptive authentication mechanisms will reduce password reuse and improve user trust, addressing gaps in current usability-focused approaches.

Objectives

1. To evaluate the extent of password reuse across multiple accounts among different demographic groups within six months.
2. To analyze the impact of password reuse on overall system vulnerability by simulating attack scenarios within a controlled environment.
3. To compare the effectiveness of password managers versus manual password strategies in reducing reuse over a three-month experimental period.
4. To identify usability limitations in current password policies that encourage reuse and propose improvements within one year.

4 Literature Review Summary

This review's goal was to examine password usability with a particular focus on the relationship between security procedures and user behavior. The primary findings show that, although identification-based methods and password managers offer some solutions, memorability, storage practices, and password reuse remain significant issues. Significant research limitations include a lack of long-term usability studies, a limited number of different user samples, and the insufficient examination of hybrid or passwordless systems. The general conclusion is that current approaches are insufficient to achieve a balance between security and usability, despite developments. Future research alternatives include developing flexible and user-friendly authentication methods, looking into hybrid models, and implementing passwordless technologies to address unresolved usability difficulties.

5 Research Methodology

5.1 Research Design

The research will use a mixed-methods approach that combines simulation-based modeling, quasi-experimental comparisons, and empirical surveys. While controlled tests will contrast the use of password managers with manual methods, surveys will record user behavior and demographic influences on password reuse. Breach probability under various reuse circumstances will be modeled using Monte Carlo simulations. This design ensures quantitative validation of security threats as well as qualitative insights into user behavior.

5.2 Technical Approach

1. Monte Carlo Simulation : Used to estimate breach likelihood when reused passwords are exposed in large datasets.

2. Comparative Framework : Two user groups, password manager users and non-users, will be studied to measure differences in password uniqueness and strength.

3. Algorithm Development : Prototype an explainable password audit tool that provides transparent feedback on password weaknesses.

4. Usability Testing : Apply heuristic evaluation and task-based testing to identify limitations in password creation policies and their effect on reuse.

5.3 Data and Tools

- Statistical Packages: SPSS, Python for quantitative analysis and modeling.

- Password Managers: Bitwarden, KeePass, LastPass for experimental comparison between manager users and non-users.

- Simulation Frameworks: MATLAB, Python (NumPy/SimPy) for Monte Carlo modeling and probabilistic simulations.

5.4 Evaluation Criteria

- Security Metrics: Password entropy, uniqueness, and resistance to dictionary/brute-force attacks.
- Usability Metrics: Task completion time, error rates, memorability, and user satisfaction scores.
- Performance Metrics: Simulation accuracy, computational efficiency, and scalability of algorithms.
- Adoption Metrics: Percentage of participants reducing reuse after exposure to password managers or audit tools.
- Explainability Metrics: User comprehension of audit feedback, measured through post-task questionnaires.

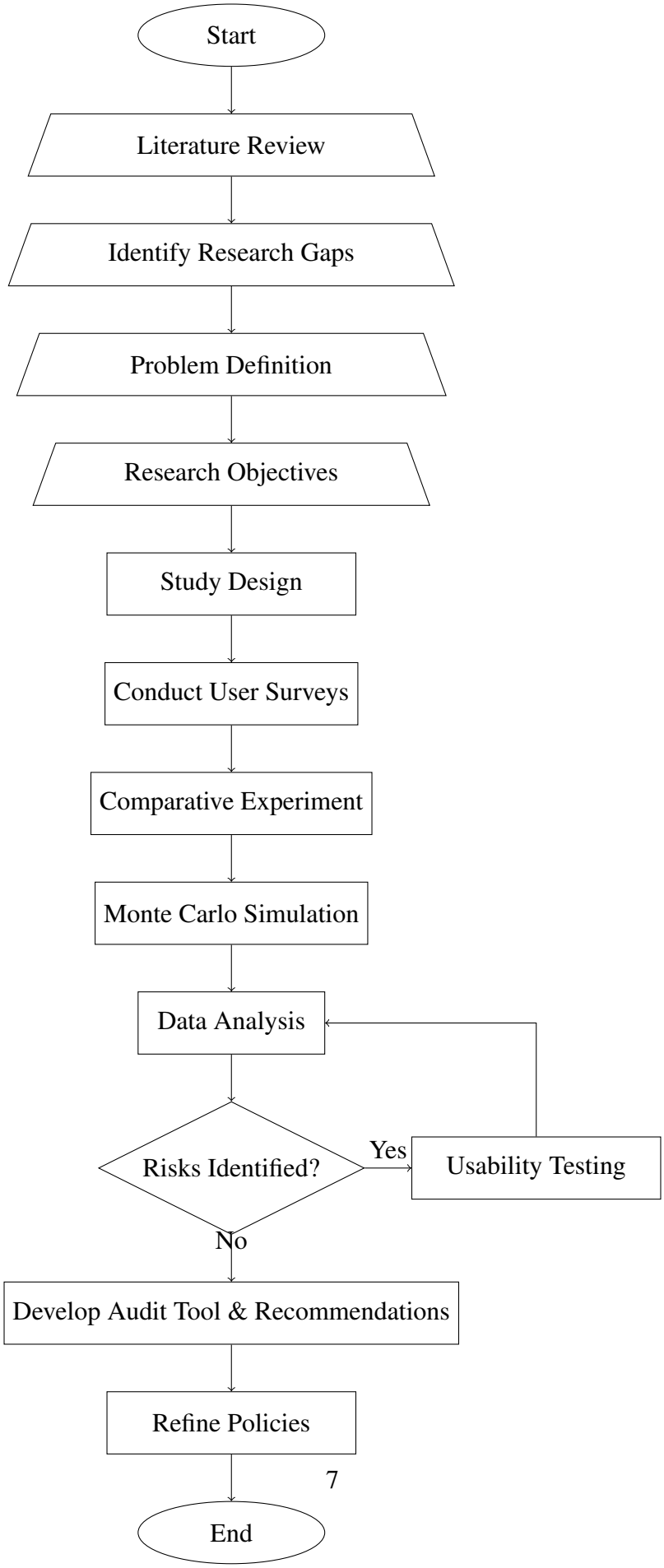
6 Expected Outcomes

The results of this study should demonstrate that password reuse is still common among a variety of user demographics, primarily due to usability issues and mental limitations. The study will probably reveal that reused passwords greatly enhance vulnerability to widespread breaches by examining survey data and simulation results. It is anticipated that comparative studies will show that password managers, when paired with credential audit tools, boost password uniqueness and decrease reuse in comparison to manual methods. By offering empirical proof of the usefulness of supporting tools in reducing insecure user behaviors, these findings will advance the discipline.

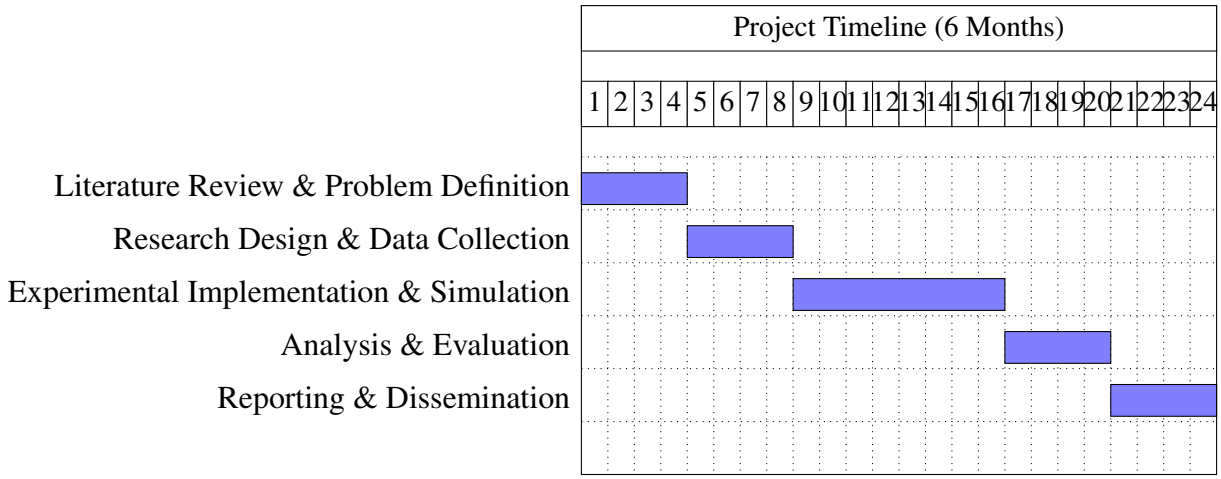
These results are important because they increase our knowledge of password usability in both academic and practical contexts. A comprehensive dataset on password reuse trends, a comparative assessment approach for password managers, and a prototype explainable audit tool that boosts user trust by offering clear feedback are among the outputs. Together, these results will influence cybersecurity policy, direct the creation of user-friendly authentication systems, and create new possibilities for the integration of security technology with usability-focused methodologies. Thus, this study will close the gap between theoretical password security models and actual user behavior.

7 Research Activities and Milestones

7.1 Research Activities Flowchart



7.2 Research Activities Gantt Table



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